Date.
Page No.

Magnetic: Reluctance
iv) Law Similarity :-
Ohm's law $\longleftrightarrow$ Magnetic circuit law

5.) $l = 40cm = 0.4m$
$A = 5cm^2 = 5\times10^{-4}m^2$
$\mu r = 1000$
$N = 500$
$I = 2A$
a) $H = \frac{NI}{l} = \frac{500\times2}{0.4}$
$= 2500 A/m$

b) $B = \mu_0\mu_r H$
$= (4\pi\times10^{-7})\times1000\times2500$
$= 3.14T$

c) $\phi = B.A$
$= 3.14\times5\times10^{-4}$
$= 1.57\times10^{-3} Wb$

6.) $\mu r = 2000$
$li = 50cm = 0.5m$
$lg = 1mm = 0.001m$
$A = 4cm^2 = 4\times10^{-4}m^2$
$\phi = 0.8mWb = 0.8\times10^{-3}$

$B = \frac{\phi}{A} = \frac{0.8\times10^{-3}}{4\times10^{-4}}$
$= 2T$
$Hi = \frac{B}{\mu_0\mu_r}$